Ref. [43] — Adam: A Method for Stochastic Optimization

D. P. Kingma, J. Ba | International Conference on Learning Representations (ICLR), 2015

1. Goal: combine momentum with per-coordinate adaptivity

Deep networks have noisy, nonstationary, and highly uneven gradients. Kingma and Ba introduce Adam (*adaptive moment estimation*), a first-order optimizer that combines an exponential average of gradients (momentum-like behavior) with an exponential average of squared gradients (coordinate-wise scaling). Their empirical contribution is the practicality of this combination; the paper's theoretical regret analysis is framed in online convex optimization.

2. Derive the algorithm step by step

At iteration t, compute stochastic gradient g_t. Maintain two running statistics:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t,$$
$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}.$$

Here $g_t^{\odot 2}$ squares coordinates separately. With $m_0 = v_0 = 0$, these moving averages are initially biased toward zero; use

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \qquad \hat{v}_t = \frac{v_t}{1-\beta_2^t}.$$

Then update each parameter:

$$\boxed{\theta_t = \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon_{\text{num}}}} \quad \text{(elementwise)}.$$

The small ε_{num} prevents division by zero; it is unrelated to the ϵ used in convergence-to-stationarity theorems.

3. Why bias correction is needed (first-principles example)

Suppose the gradient is a constant g and $m_0 = 0$. Expanding the recurrence,

$$m_1 = (1-\beta_1) g, \qquad m_2 = (1-\beta_1)(1+\beta_1) g,$$

so after t steps,

$$m_t = (1-\beta_1) \sum_{k=0}^{t-1} \beta_1^k g = (1-\beta_1^t) g.$$

Therefore $\hat{m}_t = g$. The same calculation with g^2 gives $\hat{v}_t = g^2$ for a constant gradient. Without bias correction, a zero-start moving average initially underestimates its long-run value.

4. What the paper actually establishes

The original algorithm is inexpensive and uses $O(d)$ memory. The authors give a regret analysis for the online convex setting under their stated boundedness and parameter conditions, plus experiments showing competitive training behavior. This is *not* a generic theorem that Adam converges to a global minimum of a modern non-convex neural network; subsequent research identified limitations of the original convergence argument and counterexamples for Adam-type updates under certain settings.

Main contribution. A practical adaptive optimizer that uses bias-corrected first and second gradient moments to combine directional persistence with coordinate-wise rescaling.

5. Contrast with AdaGrad and SGD

SGD uses $\theta_t = \theta_{t-1} - \alpha g_t$ (or a velocity update with momentum). AdaGrad divides by a cumulative sum of squared gradients. Adam's exponential averages *forget old gradients*, which can adapt rapidly but also means that convergence arguments do not automatically carry over from AdaGrad. The ratio $\hat{m}_t/(\sqrt{\hat{v}_t} + \varepsilon_{\text{num}})$ is a *preconditioned direction*, so stability should account for the optimizer-induced metric, not just the Euclidean Hessian.

6. Limits and survey connection

The paper does not establish a finite-time ϵ-SOSP guarantee, explicit negative-curvature search, or strict-saddle escape for ordinary Adam. Its online regret bound has assumptions different from later non-convex results, such as our Ref. [34]. We use it for **RQ2** (modern training dynamics) and **RQ4** (what remains unproved about adaptive large-step training).

> **Accuracy note.** "Adam works well in experiments" and "Adam always converges on neural-network objectives" are different claims. The original paper supports the former, not the latter.

7. Our notes

This paper explains the optimizer used in so much modern deep learning: one moving average remembers the direction of gradients, while another rescales each coordinate by recent gradient magnitudes. Bias correction is important at the start. For our survey, Adam is evidence of how real optimization differs from textbook GD, not a proof of saddle escape. **Original source:** `https://arxiv.org/abs/1412.6980`. **Source check:** original author manuscript and ICLR publication metadata.